

Counterfactual Effect Binding for LLM-Agent Tool Safety

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Abstract

LLM-agent safety monitors must reason about realized effects, affected resources, authorization context, provenance, and utility rather than only tool names or trace formats. This paper package reports controlled custom-stress evidence for an Effect-Binding Guard and a local pre-commit authorization prototype. Existing results show that hard tuple binding can reduce unsafe pre-allow under custom stress, but E50 exposes a resource/authorization bottleneck. Finally, E55, a local mock pre-commit mediation study, shows that typed authorization context and atom-level action expansion can turn the old guard’s abstention-heavy behavior (coverage 0.200) into high-coverage controlled decisions (coverage 0.920), while preserving zero observed unsafe pre-allow in the constructed setting. This is contract evidence under explicit authorization infrastructure, not a production safety certificate.

1 Introduction

LLM agents increasingly mediate side-effectful tools such as messaging, file sharing, calendars, workspace administration, and transaction APIs. Safety decisions for these tools cannot be reduced to a tool name. A single tool surface may realize different effects, and the same effect may appear through different tools, wrappers, or trace formats.

Prior experiments in this repository show intentionally mixed evidence. Counterfactual stress tests identify tool-effect fragmentation and structural defenses that remain sensitive to schema or graph surfaces. The hard Effect-Binding Guard is therefore framed as a measurement and method-feasibility artifact: it binds candidate actions into effect, resource, authorization, and provenance tuples, then uses disagreement, evidence fallback, and provenance checks to decide whether to allow, deny, or abstain.

Reviewer-facing interpretation. E55 shows that the resource/authorization bottleneck is also an interface problem. Without explicit authorization infrastructure, the old hard guard abstains heavily. With typed authorization context, resource alias maps, operation-mode semantics, multi-resource atom expansion, and provenance metadata, many previously unresolved pre-commit decisions become decidable in a controlled local mock setting.

2 Threat Model and Security Goal

We study pre-commit mediation for LLM-agent tool calls. The monitor observes a user task, visible tool schema, candidate tool call, non-oracle evidence summaries when available, and provenance/control metadata. It must decide before commit whether the realized effects are authorized by the task.

2.1 Authorization Contract

E55 assumes an explicit authorization contract: authorized resource sets, aliases, operation modes, multi-resource expansion rules, public/private visibility constraints, trusted and untrusted control sources, and evidence availability. The guard may abstain when resource identity, provenance, or evidence is insufficient. This contract is visible infrastructure, not a hidden label.

3 Method

The reference hard Effect-Binding Guard predicts a tuple consisting of realized effect, resource, authorization match, and provenance risk. It is useful as a binding monitor but remains conservative when resource identity or authorization scope is underspecified.

The E55 authorization-aware prototype adds a pre-commit atom layer. A visible tool call is expanded into effect-resource-operation atoms, one per recipient, attendee, file, channel, account, payee, visibility, or operation component. The default policy requires every atom to be authorized. Alias resolution, operation-mode handling, provenance overlay, and optional evidence fallback are explicit infrastructure modules. The prototype never uses expected decisions, gold expanded atoms, violation reasons, or scoring metadata in strict mode.

4 Results

Finding 6: Explicit authorization infrastructure turns abstention-heavy binding into high-coverage local pre-commit mediation. On 600 local mock pre-commit rows, the existing hard guard has UPA 0.037, FDeny 0.000, and coverage 0.200. The authorization-aware guard has UPA 0.000, FDeny 0.000, and coverage 0.920.

The ablations indicate why the infrastructure matters. Removing multi-resource expansion raises UPA to 0.333. Removing operation-mode semantics raises UPA to 0.148. Removing provenance overlay raises UPA to 0.148. Removing alias resolution leaves UPA at 0.000 but creates FDeny 0.211. Strict replay matches original deployable metrics: True. The decision-path audit passes: True.

This finding is intentionally scoped. E55 is controlled local mock contract evidence; it does not execute real external side effects and does not validate production SaaS, browser, email, or banking systems.

5 Limitations

E55 shares deterministic mock schemas and authorization contracts with the label generator. Therefore it is contract evidence rather than independent deployment generalization. Real deployment still requires independent trace collection, a policy user interface, robust resource identity, mediation and commit protocols, logging, and adversarial validation against adapted tools. The negative E50 resource/auth stress remains an important limitation and motivation.

6 Conclusion

The current evidence supports a measurement and diagnostic framework plus a local pre-commit authorization prototype. It does not support a complete defense or production-ready guard. The main lesson is that effect binding requires explicit interfaces for authorization, resources, provenance, operation mode, and evidence.

A E55 Appendix

Method	UPA	FDeny	Coverage	Atom exact
existing_hard_effect_binding_guard	0.037037037037037035	0.0	0.2	0.0
authz_aware_effect_binding_guard	0.0	0.0	0.92	1.0
authz_aware_no_alias_resolution	0.0	0.21052631578947367	0.92	1.0
authz_aware_no_multi_resource_expansion	0.3333333333333333	0.0	0.92	0.3
authz_aware_no_operation_mode	0.14814814814814814	0.0	0.92	1.0
authz_aware_no_provenance_overlay	0.14814814814814814	0.0	0.94	1.0

Table 1: E55 strict label-hidden ablation summary.

Decision-path audit passed: True. Strict replay metric equality: True. Claim boundary: controlled local pre-commit contract evidence only.